

Algebra: Adding & Subtracting Rational Expressions

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DIRECTIONS

Add or subtract the rational expressions. Find the LCD when denominators differ, rewrite each fraction, then combine and simplify.

1. Add: $\frac{3}{x} + \frac{5}{x}$ Answer: _____	2. Subtract: $\frac{7}{(2a)} - \frac{3}{(2a)}$ Answer: _____
3. Add: $\frac{2}{(x+1)} + \frac{4}{(x+1)}$ Answer: _____	4. Subtract: $\frac{5}{(y-3)} - \frac{2}{(y-3)}$ Answer: _____
5. Add: $\frac{1}{x} + \frac{1}{y}$ Answer: _____	6. Add: $\frac{2}{(3x)} + \frac{1}{x}$ Answer: _____
7. Subtract: $\frac{5}{(6a)} - \frac{1}{(4a)}$ Answer: _____	8. Add: $\frac{3}{(4x)} + \frac{2}{(3x)}$ Answer: _____
9. Add: $\frac{2}{(x+2)} + \frac{3}{(x-1)}$ Answer: _____	10. Subtract: $\frac{4}{(x+3)} - \frac{1}{(x-2)}$ Answer: _____

Based on the Numberbender lesson "ALGEBRA: Adding and Subtracting Rational Expressions" • youtu.be/24iEx_9wSUQ

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Answer Key & Solutions

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TEACHER NOTES

Items 1-4: same denominator — combine numerators directly. Items 5-8: different monomial denominators — find LCD. Items 9-10: binomial denominators — factor before finding LCD.

1. Add: $3/x + 5/x$

Answer: $8/x$

Same denominator; $3+5=8$, over x .

2. Subtract: $7/(2a) - 3/(2a)$

Answer: $4/(2a) = 2/a$

$7-3=4$ over $2a$; simplify $\rightarrow 2/a$.

3. Add: $2/(x+1) + 4/(x+1)$

Answer: $6/(x+1)$

$2+4=6$, over $(x+1)$.

4. Subtract: $5/(y-3) - 2/(y-3)$

Answer: $3/(y-3)$

$5-2=3$, over $(y-3)$.

5. Add: $1/x + 1/y$

Answer: $(x+y)/(xy)$

$LCD = xy$; $y/(xy) + x/(xy) = (x+y)/(xy)$.

6. Add: $2/(3x) + 1/x$

Answer: $5/(3x)$

$LCD = 3x$; $2/(3x) + 3/(3x) = 5/(3x)$.

7. Subtract: $5/(6a) - 1/(4a)$

Answer: $7/(12a)$

$LCD = 12a$; $10/(12a) - 3/(12a) = 7/(12a)$.

8. Add: $3/(4x) + 2/(3x)$

Answer: $17/(12x)$

$LCD = 12x$; $9/(12x) + 8/(12x) = 17/(12x)$.

9. Add: $2/(x+2) + 3/(x-1)$

Answer: $(5x+4)/[(x+2)(x-1)]$

$LCD=(x+2)(x-1)$; $2(x-1)+3(x+2)=5x+4$.

10. Subtract: $4/(x+3) - 1/(x-2)$

Answer: $(3x-11)/[(x+3)(x-2)]$

$LCD=(x+3)(x-2)$; $4(x-2)-1(x+3)=3x-11$.

